MIL-A-45872(PA)
9 June 1975
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

ARM-SAFE DEVICE ATOMIC WEAPON M38 ASSEMBLY, PACKING, AND MARKING

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of training arming-safing mechanism entitled as Arm-Safe Device, Atomic Weapon M38, which is used with the Nike Hercules atomic warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-10687	- Enamel, Lusterless, Quick-Drying (for Ammunition).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1167	- Ammunition Data Card.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

ORDNANCE CORPS

8785755	- Base Assembly.
8785758	- Arming-Safety Mechanism T-105.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The arm-safe device XM38 shall comply with all requirements specified on the drawing (dwg) 8785758, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Coating. All exterior surfaces shall be painted with blue No. 3530 Enamel in accordance with MIL-E-10687.

3.5 Marking. Each arm-safe device XM38 shall be marked "For Training Only" using white No. 3715 Enamel, in accordance with MIL-E-10687.

3.6 Insulation resistance. The insulation resistance shall be 100 megohms minimum when tested at 500 ± 50 volts dc between pins, and between all pins and the case except for pins electrically connected when tested as specified in 4.4.1.

3.7 Continuity. The XM38 arm-safe device shall show continuity using dwg 8785755 and information listed in this specification when tested as specified in 4.4.2.

3.8 Functioning. The assembly, when subjected to accelerations of 12 ± 3 to 25 ± 3 g to sustained for a minimum of 2.5 seconds in the direction of forward motion, shall actuate its timing mechanism which shall then close the normally open electrical contacts within a period of 1.7 to 2.5 seconds when tested as specified in 4.4.3.

3.9 Workmanship.

3.9.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.9.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.9.3 Burrs. Parts shall be free from burrs.

3.9.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.9.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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3.9.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8785758) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Insulation resistance (see 3.6)		X
3. Continuity (see 3.7)		X
4. Functioning (see 3.8)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

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4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

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4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Base assembly prior to applying embedding compound (see dwg 8785755).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Wire loose or improperly connected	Visual/Manual
Minor:		
	201. Evidence of poor workmanship (see 3.9)	Visual

4.3.2.1.2 Base assembly (see dwgs 8785755 and 8785758).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Switch broken or loose	Visual/Manual
	102. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Connector key mislocated	Visual
	202. Embedding compound missing or inadequate	Visual
	203. Evidence of poor workmanship (see 3.9)	Visual
	204. Hardware loose or missing	Visual/Manual
	205. Pin loose or missing	Visual/Manual

4.3.2.1.3 M38 Arm-safe device, prior to assembling cover (see dwg 8785755).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Lubricant missing	Visual
	102. Component loose or missing	Visual/Manual

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Minor:
201. Evidence of poor workmanship (see 3.9).....Visual

4.3.2.1.4 M38 Arm-safe device (see dwg 8785758).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Diameter of base mounting holes	Gage
102.	Location of base mounting holes	Gage
103.	Not set on "Safe"	Visual
104.	Component loose or missing	Visual/Manual
Minor:		
201.	Evidence of poor workmanship (see 3.9)	Visual
202.	Marking missing, misleading, or illegible	Visual

4.3.2.1.5 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packing, and packaging not in accordance with MIL-STD-1169	Visual
102.	Marking not in accordance with MIL-STD-1169	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.9)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test classification
a.	Insulation resistance	Major
b.	Continuity	Major
c.	Functioning	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

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4.3.3.1 Insulation resistance continuity and functioning. Each arm-safe mechanism shall be tested in accordance with 4.4.1, 4.4.2 and 4.4.3. If any item fails to comply with 3.6, 3.7 and 3.8, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major and Minor characteristics and tests. Inspection and testing shall be performed with the acceptance test equipment, operating instructions and calibration instructions prepared by the contractor. The contractor shall obtain approval of such designs, operating instructions and special calibration instructions as specified in 6.3 prior to use on the contract.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Insulation resistance. The insulation resistance between pins of the receptacle, and between pins and the base, shall be more than 100 megohms using an applied voltage of 500 ± 50 volts dc; see dwg 8785755 and information listed below.

Each pin to base assembly.

A to B, C, D, E, F, H, J.
B to C, D, E, F, H, J.
C to D, E, F, H, J.
D to E, F, H, J.
E to F, H, J.
F to H, J.
H to J.

4.4.2 Continuity. Each conductor between the receptacle pin and its appropriate switch contact terminal shall be tested for continuity, using dwg 8785755 and information listed below. An ohmmeter or other suitable electrical testing device may be used.

From Receptacle Pin	to	Switch Assembly
A	to	Ground
B	to	Open no connection
C	to	S1 (normally open)
D	to	S1 (common)
E	to	S1 (normally closed)
F	to	S2 (normally open)
H	to	S2 (common)
J and K	to	Connected to each other
L	to	S2 (normally closed)

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4.4.3 Functioning. The assembly shall be mounted on a centrifuge with its cover removed. A suitable trigger mechanism, which can be released electrically in series with an electrical timer, shall be installed on the centrifuge to hold the acceleration mass of the assembly from moving until the desired g level is reached. A snap action table on the centrifuge may be used at discretion of the contractor. If this method is used, the cover shall be left on the assembly and the release connected in series to the timer.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The arm-safe device shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Level C. The arm-safe device shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.2 Packing.

5.2.1 Levels A and C. The arm-safe device shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Data cards. Data cards shall be prepared in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 and 5.4 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

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6.4 Supersession data. This specification supersedes the requirements of Purchase Description X-PA-PD-1515 dated 25 March 1959.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A188

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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